

A decorative graphic on the left side of the slide. It consists of a blue parallelogram and a light green parallelogram, both tilted at an angle. The blue shape is in the foreground, and the green shape is partially behind it. They are set against a dark blue background with subtle diagonal lines.

Bank Marketing Targeting Optimization



Business Problem:

Profit-Oriented Customer Targeting in Bank Marketing

- The bank is running a term-deposit marketing campaign.
- Contacting every customer is costly.
- A customer who subscribes may bring deposit funding to the bank.
- That deposit generates economic value, but outreach and servicing also create costs.



Decision Problem

Decision Problem

Given a limited campaign budget, which customers should the bank target in order to maximize expected profit?

Why This Matters

- Improve campaign conversion rate
- Reduce unnecessary contact cost
- Increase marketing ROI and profitability



Data Source and Initial EDA

DATASET OVERVIEW:

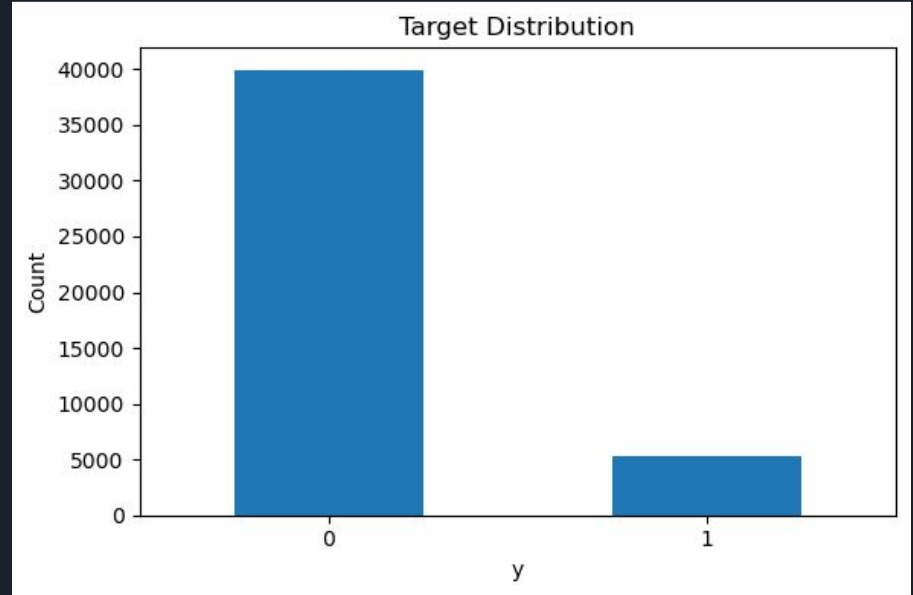
- Bank Marketing dataset (45,211 observations, 16 features)
- Binary target: term deposit subscription ($y = 1 / 0$)
- Features include demographics, financial status, and campaign interactions

DATA QUALITY:

- Significant missing values in categorical variables
- outcome (~82%), contact (~29%), education and job (minor)
- Requires imputation and careful preprocessing

Target Distribution

- No subscription: 39,922 (88.3%)
- Subscription: 5,289 (11.7%)
- Highly imbalanced classification problem
- Need metrics beyond accuracy (e.g., recall, AUC)





Estimation Modeling Plan

- **Prediction task:** predict whether a client will subscribe to a term deposit.
- **Target variable y :**

$$y_i = \begin{cases} 1, & \text{if client subscribes to a term deposit} \\ 0, & \text{otherwise} \end{cases}$$

Independent variables (X):

- **Customer profile:** age, job, marital status, education
- **Financial status:** default, balance, housing loan, personal loan
- **Campaign/contact variables:** contact type, day of month, month, duration, campaign
- **Previous campaign history:** pdays, previous, poutcome



Decision Optimization Plan: Budget-Constrained Customer Targeting

Optimization Goal

Select the subset of customers that **maximizes expected campaign profit** under a fixed marketing budget.

Objective Function

$$\max \sum_i z_i [\hat{p}_i V_i - c_{\text{contact}}]$$

Customer Value

$$V_i = A_i \cdot s \cdot T - c_{\text{service}} - c_{\text{promo}}$$

Constraint

$$\sum_i z_i c_{\text{contact}} \leq B, \quad z_i \in \{0, 1\}$$

Decision Variables & Parameters

$i = 1, 2, \dots, N$: customer index

$z_i \in \{0, 1\}$: whether customer i is targeted

p_i : predicted probability that customer i subscribes

V_i : net economic value generated if customer i successfully subscribes

c_{contact} : cost of contacting one customer

A_i : amount of money the customer will deposit

s : bank's annual net financial spread earned on deposit funding

T : effective holding horizon

c_{service} : cost of operation

c_{promo} : cost of promotion

B : total campaign budget



Baseline Assumptions

- Deposit amount assumption:
 - $A_i = 2000$ if $balance_i \leq 0$
 - $A_i = 4000$ if $0 < balance_i \leq 1000$
 - $A_i = 6000$ if $1000 < balance_i \leq 5000$
 - $A_i = 8000$ if $balance_i > 5000$
- Financial spread assumption: $s = 3.57\%$
- Effective holding horizon: $T = 1.0$ year
- Service cost: $c_{\text{service}} = 5$
- Promotional cost: $c_{\text{promo}} = 0$
- Contact cost: $c_{\text{contact}} = 10$
- Campaign budget: $B = 30,000$

Traditional Statistical Model: Weighted Logistic Regression

Top positive signals:

- poutcome_success
- duration
- month_mar, month_oct, month_sep
- job_student, job_retired
- housing_no, loan_no

Top negative signals:

- poutcome_failure
- campaign
- housing_yes
- month_may, month_jan, month_jul

Top Positive Coefficients		Top Negative Coefficients	
Feature	Coef.	Feature	Coef.
month_mar	1.8108	job_entrepreneur	-0.2258
poutcome_success	1.6465	education_primary	-0.2458
duration	1.4945	job_blue-collar	-0.2576
month_oct	1.2766	campaign	-0.2859
month_sep	0.9613	job_self-employed	-0.2993
month_dec	0.8961	job_housemaid	-0.3420
job_student	0.7218	housing_yes	-0.3852
job_retired	0.5981	month_jun	-0.5169
housing_no	0.5013	poutcome_other	-0.5768
loan_no	0.3414	month_aug	-0.7386
education_tertiary	0.3258	month_nov	-0.7970
month_apr	0.2423	month_jul	-0.8657
marital_single	0.2126	poutcome_failure	-0.9536
job_admin.	0.1909	month_jan	-1.0932
default_no	0.1632	month_may	-1.1074

These results are intuitive and provide business insight into customer response behavior.

This makes weighted logistic regression a useful benchmark for explaining targeting decisions.



Machine Learning Models: Random Forest and XGBoost

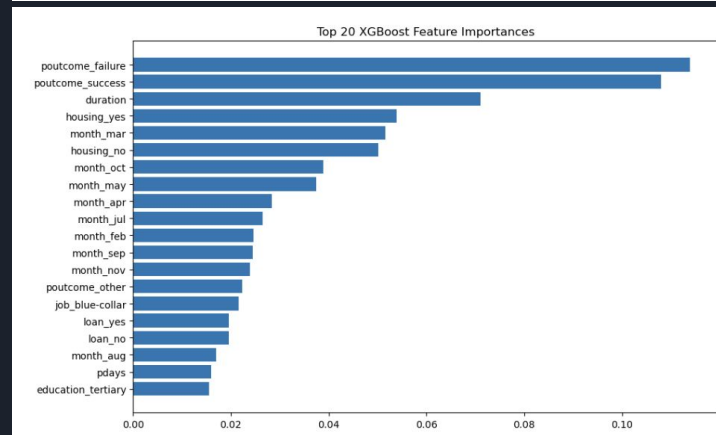
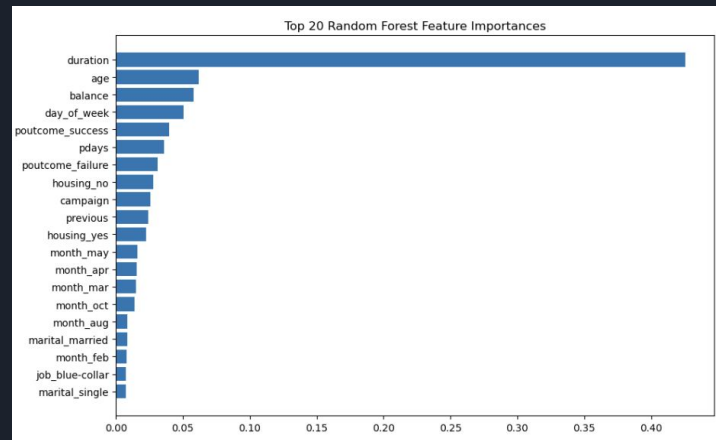
- We split the data into:
 - **80% training set**
 - **20% test set**
- The split is **stratified** to preserve the class ratio.
- **In-sample validation:** evaluate model performance on the training set
- **Out-of-sample validation:** evaluate model performance on the test set

Evaluation metrics:

- Accuracy
- Precision
- Recall
- F1-score
- AUC

Interpretation of Random Forest and XGBoost

- RF emphasizes **duration**, **age**, **balance**, and **previous campaign signals**
- XGBoost emphasizes **poutcome**, **duration**, and **month-related timing effects**
- Both models consistently identify **campaign history and contact variables** as strong predictors





Preliminary Model Performance

Because the dataset is imbalanced, we place particular emphasis on **Recall**, **F1-score**, and **AUC**.

Dataset	Model	Accuracy	Precision	Recall	F1-score	AUC
Train	Weighted Logistic Regression	0.8415	0.4099	0.8069	0.5436	0.9057
Test	Weighted Logistic Regression	0.8387	0.4029	0.7864	0.5328	0.9020
Train	Random Forest	0.9088	0.5660	0.9461	0.7082	0.9784
Test	Random Forest	0.8775	0.4852	0.7769	0.5974	0.9224
Train	XGBoost	0.8714	0.4745	0.9241	0.6270	0.9564
Test	XGBoost	0.8555	0.4390	0.8469	0.5783	0.9263

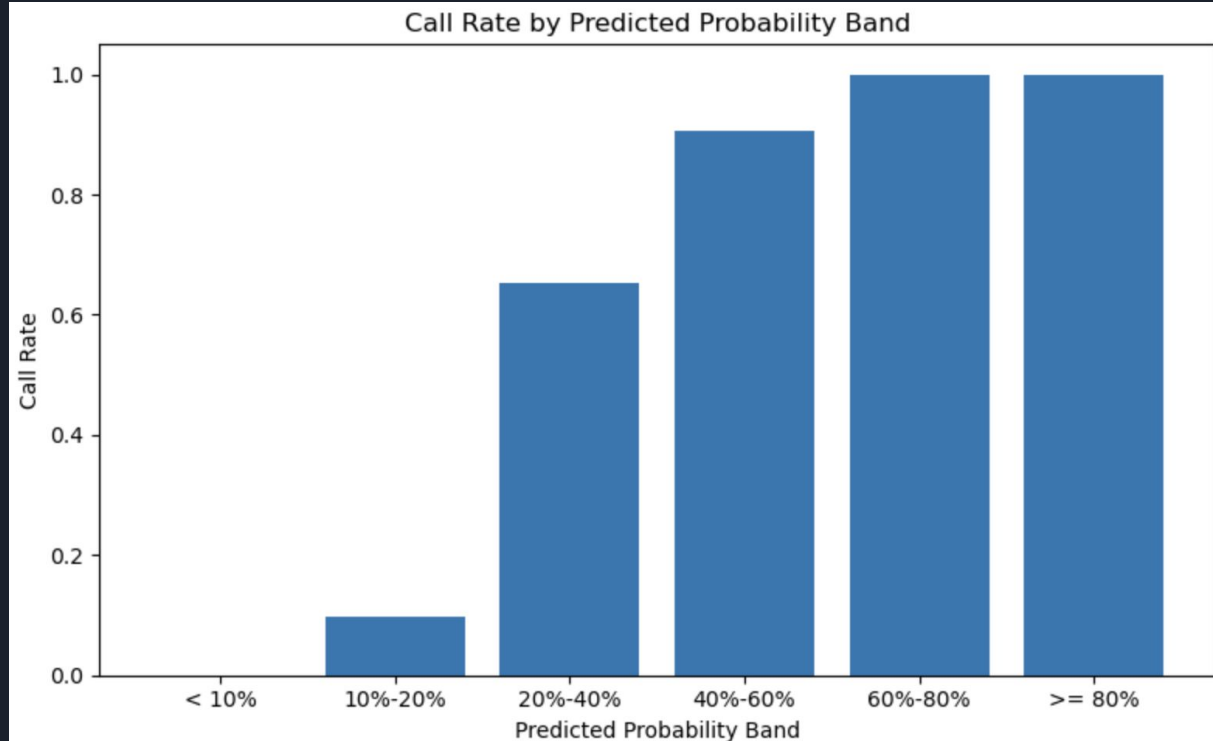
- All three models perform well in distinguishing likely subscribers from non-subscribers.
- XGBoost achieves the highest AUC and recall.
- Random Forest achieves the strongest F1-score and the highest accuracy.
- Weighted Logistic Regression remains useful because of its interpretability.



Optimization Solution

	Metric	Value
	Total customers in candidate set	9043
	Customers with positive expected profit	5649
	Maximum calls allowed by budget	3000
	Customers actually selected to call	3000
	Customers not selected	6043
	Average expected profit of called customers	84.08
	Average expected profit of not-called customers	1.44
	Expected total profit of called customers	252240.45
	Actual total profit of called customers	137602.4
	Actual conversion rate among called customers	0.3267
	Actual conversion rate overall	0.117

Optimization Solution





Business implications

- The bank cannot call everyone because the campaign budget is limited.
- Therefore, the key business problem is how to prioritize customers under fixed outbound capacity.
- The optimization framework produces a ranked call list based on expected economic value.
- This improves return per call and makes campaign execution more efficient.



Recommendations and Implementation

Optimal Strategy

- Deploy XGBoost as the primary targeting model.
- Rank customers by expected profit score .
- Apply budget-constrained optimization to generate a prioritized call list.
- Allocate campaign budget toward highest expected-profit customers only.

Implementation Guidelines

- Exclude post-contact variables for realistic pre-call targeting.
- Replace constant value assumptions with customer-specific deposit estimates.
- Add operational constraints to the optimizer.
- Conduct sensitivity analysis on budget level and deposit spread rate.



Conclusion

Key Takeaways

- Bank marketing is reframed as a profit optimization problem, not just classification.
- XGBoost achieves the best AUC and recall — recommended for deployment.
- The optimization framework translates predictions into ROI-maximizing targeting decisions.

Strengths and Limitations

- End-to-end pipeline from raw data to business decision.
- Economically grounded objective function with explicit profit modeling.
- Duration is post-contact, reducing real-world pre-call applicability.
- Revenue and cost currently assumed constant across all customers.
- Customers treated independently; no interaction or network effects modeled.

Future Improvements

- Build restricted model excluding post-contact variables.
- Incorporate customer-specific deposit value estimates.
- Extend optimization with richer constraints and sensitivity analysis.